

TradeVed Stress Tester — Market Research & Product Strategy

Purpose: A build-ready product strategy document for evolving the TradeVed Stress Tester into a best-in-market strategy-robustness product.

Audience: Product, quant research, and engineering.

Date: May 2026

Method: Competitive teardown of 20+ platforms, the academic overfitting literature, institutional risk practice, and a line-level read of TradeVed's current `engine/stress.py`.

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1. Executive Summary

The stress-testing / strategy-robustness market splits into **three tiers**, and there is a **wide-open gap in the middle** that TradeVed is unusually well positioned to own.

Tier	Who	What they do	Price	The catch
Institutional	BlackRock Aladdin, Bloomberg PORT, MSCI RiskMetrics, FactSet	Portfolio-level factor/macro/climate stress, correlation-breakdown modeling, historical replay with <i>your exposures</i>	\$100K– \$1M+/yr	No retail access, no algo-strategy testing, no self-serve
Quant/Robustness	Build Alpha, StrategyQuant X, AmiBroker, vectorbt	Deep robustness suites: Monte Carlo (5–9 variants), walk-forward, noise/synthetic data, parameter sensitivity, overfitting rejection	\$99–\$1,290 one-time or sub	Desktop-bound, steep learning curve, no Indian market, ugly tables not visuals, no streaming/live UX
Retail/Charting	TradingView, TrendSpider, NinjaTrader, AlgoTest, Streak	Easy backtests, great charts, broker integration	Free– \$60/mo	Backtest <i>stops at the backtest</i> — little/no real stress testing, no overfitting science, no robustness score

The strategic insight: Nobody combines (a) institutional-grade stress methodology, (b) the robustness/overfitting science quants trust, (c) a beautiful real-time retail UX, and (d) Indian + crypto + US market coverage. Each tier is missing two of the four.

TradeVed already has rare assets that the retail and even quant tiers lack:

- **Scenario-based pre-strategy OHLCV perturbation** (13 historical presets) — the strategy reacts to the shock *naturally*, which is methodologically closer to Build Alpha's "synthetic data stress" than to the cheap "shuffle the trade list" Monte Carlo most retail tools ship.
- **Live SSE-streamed Monte Carlo** with a **canvas spaghetti chart** (1,000+ paths) and a **delta-view toggle** — a genuinely novel, institutional-feeling UX that *no retail competitor has*.
- **Per-run magnitude jitter** (`severity × uniform(0.75, 1.25)`) so paths fan out in both timing and intensity.
- **Indian cost model** (STT/GST/SEBI/stamp, Budget 2024) + F&O lot sizes — a moat in the fastest-growing retail algo market on earth.

Important nuance (verified in code): TradeVed *already ships* a full **walk-forward + out-of-sample validation engine** (`engine/validation.py` → `run_holdout` , `run_walk_forward`), surfaced via `ValidationPanel.tsx` . But it lives in the **backtester** flow only — the **stress tester** does not call it. So the gap is *integration + scoring*, not building walk-forward from scratch.

What the stress tester is still missing is the *validation science* that turns a pretty stress chart into a **trustworthy verdict**: the existing walk-forward/OOS engine isn't wired in, and there's no overfitting probability (PBO), no Deflated Sharpe, no Monte Carlo permutation p-value, no parameter-sensitivity heatmap, no single robustness score, no tail-risk metrics (CVaR/probability of ruin), and only single-asset coverage (no portfolio correlation-breakdown).

The recommendation in one line: Keep the world-class scenario engine and streaming UX, and bolt on a **Validation & Confidence layer** — a single **Robustness Score (A+→F)** backed by walk-forward, an overfitting probability, tail-risk metrics, and a parameter-sensitivity heatmap — so TradeVed becomes the only product that *stress-tests a strategy AND tells you whether to trust the backtest at all*, in real time, for Indian/crypto/US markets, with a UI a retail trader actually enjoys.

2. Competitor Comparison Table

Platform	Core offering	Target user	Pricing (2026)	Killer feature	Biggest weakness for stress/robustness
BlackRock Aladdin	Institutional risk OS	Asset managers, banks	\$100K–\$1M+/yr	Historical replay using <i>current exposures</i> + correlation-breakdown; macro & climate scenarios	No retail, no algo-strategy backtesting, no self-serve
Bloomberg PORT	Terminal portfolio & risk	Buy-side PMs	~\$24K/yr (terminal)	4 scenario engines: factor, full-valuation, macro, climate; build custom scenarios (even "Spanish flu")	Portfolio-level only; no strategy robustness; terminal lock-in
MSCI RiskMetrics	Factor risk & stress	Institutions	\$\$\$	"Broken-arrow" correlation-breakdown stress test; factor stress	Institutional only; not strategy-centric
HiddenLevers (Orion)	Advisory scenario analysis	RIAs/advisors	~\$300+/mo/advisor	"Scenario priced-in" indicator (key-lever tracking); macro-linked scenarios; client-ready reports	No algo testing; no per-trade MC; managed-portfolio framing
Portfolio Visualizer	Allocation backtest + MC	Retail/RIA	Free–\$360/yr	Multiple MC models (historical bootstrap / parametric / custom dist), withdrawal survival	No strategy-execution testing; portfolio-level; no scenario shocks
Build Alpha	Strategy generation + robustness	Serious retail quants	~\$999–\$1,990	The deepest retail robustness suite: noise tests, MC permutation, variance testing, vs-random, walk-forward, synthetic-data stress with custom rules	Desktop, EasyLanguage-centric, no Indian market, table-heavy UI, no live streaming
StrategyQuant X	Auto strategy builder	Quants	~€690–€1,290	Robustness as a build gate — 5+ MC methods auto-reject fragile strategies; randomize params / trade order / data	Complex; desktop; FX/futures-centric; no beautiful reporting
AmiBroker	Fast backtester	Power retail	~\$279–\$339	Fastest engine; probability of ruin; automated walk-forward (IS+OOS); random trade/price perturbation	Windows-only; no scenario library; MC output is a table; no Indian costs
QuantConnect	Cloud algo platform	Coders/quants	Free–\$60+/mo	Parameter-sensitivity heatmaps on cloud compute; point-in-time data (no lookahead); regime filtering	No stress-scenario presets; no MC spaghetti; code-required

Platform	Core offering	Target user	Pricing (2026)	Killer feature	Biggest weakness for stress/robustness
AlgoTest	No-code Indian options	Indian retail	₹0 (25/wk) – ₹499+/mo	Tick-level multi-leg options backtest; realistic fills; AI agent "920" (natural-language backtest)	No stress scenarios; no robustness/overfitting science; India-only
Streak (Zerodha)	No-code Indian algo	Indian retail	~₹500–₹900/mo	Dead-simple no-code; tight Zerodha integration	Shallow backtest; no stress testing at all
TradingView	Charting + Pine backtest	Everyone	Free–\$60/mo	Best visual feedback — signals overlaid on chart; huge community	"Deep Backtesting" is paywalled & shallow; no MC, no walk-forward, no robustness
TrendSpider	AI charting + bots	Retail	~\$30–\$120/mo	AI Strategy Lab (natural-language strategy build), 50 yrs data, automation	No slippage/commission modeling; no MC/walk-forward; no stress
NinjaTrader	Futures platform	Futures retail	Free–\$1,499	Walk-forward + MC + genetic optimizer built-in	Futures-centric; MC is basic; dated UX
MultiCharts	Pro charting/backtest	Pro retail	~\$1,497	Exhaustive + genetic optimization, transparent	No built-in walk-forward (workarounds needed); steep
Wealth-Lab	.NET backtester	Coders	~\$59/mo	Built-in walk-forward + MC + parameter-stability	.NET/C# required; niche
MetaTrader 5	Retail FX/CFD	FX retail	Free	Genetic optimization + forward testing (OOS); multi-currency	EA/MQL5 required; no scenario stress; no robustness score
TradeZella / Tradewell	Journal + backtest analytics	Retail	~\$29/mo	Closes the loop : backtest → live → analytics dashboard	Not a stress tester; no MC/scenarios
BacktestBase	MC robustness scoring	Retail	\$	30-point robustness score (A–F) ; one-click MC with 10% trade-skip; regime-layered randomization	Narrow; not a full backtester
LuxAlgo	TradingView AI overlay	Retail	~\$40/mo	Crisis templates, AI backtesting assistant, slippage modeling	Overlay, not a standalone engine
vectorbt (PRO)	Vectorized engine	Python quants	Free / €€ PRO	Insane speed (1M orders ~70ms); CPCV, purged CV, walk-forward, noise injection	Library, not a product; code-required; no UI

3. Feature Matrix

Legend: ✔ strong · ● partial/basic · ✖ absent · TV = TradeVed (today)

Capability	TV	Build Alpha	StratQuant X	AmiBroker	QuantConnect	Portfolio Viz	Aladdin	AlgoTest	TradingView
Scenario library (historical crises)	✓ 13	●	●	✗	✗	●	✓	✗	●
Pre-strategy data perturbation	✓	✓	✓	●	✗	✗	✓	✗	✗
Monte Carlo (trade reshuffle/resample)	✗	✓	✓	✓	✗	✓	✗	✗	✗
Monte Carlo (data/path)	✓	✓	✓	●	✗	✓	✓	✗	✗
MC permutation p-value	✗	✓	✓	●	✗	✗	✗	✗	✗
Walk-forward optimization	✓ [†]	✓	✓	✓	●	✗	✗	✗	✗
Out-of-sample split	✓ [†]	✓	✓	✓	✓	✗	✗	●	✗
CPCV / purged CV	✗	✗	✗	✗	●	✗	✗	✗	✗
Overfitting probability (PBO)	✗	●	●	✗	✗	✗	✗	✗	✗
Deflated / selection-bias Sharpe	✗	✗	✗	✗	✗	✗	●	✗	✗
Parameter sensitivity heatmap	✗	✓ 3D	●	✓	✓	✗	✗	✗	✗
Probability of ruin	✗	✓	●	✓	✗	●	✗	✗	✗
CVaR / Expected Shortfall	✗	●	✗	✗	●	●	✓	✗	✗
Slippage / liquidity stress	●	✓	✓	●	✓	✗	✓	●	✗
Delayed-fill / execution stress	✗	✓	✓	●	✓	✗	✓	●	✗
Correlation-breakdown / portfolio	✗	✗	✗	✗	●	●	✓	✗	✗
Reverse stress testing	✗	✗	✗	✗	✗	✗	●	✗	✗
Single robustness score (A–F)	✗	●	●	✗	✗	✗	✗	✗	✗
Live streaming MC visualization	✓	✗	✗	✗	✗	✗	✗	✗	✗

Capability	TV	Build Alpha	StratQuant X	AmiBroker	QuantConnect	Portfolio Viz	Aladdin	AlgoTest	TradingView
Canvas spaghetti (1000+ paths)	✓	✗	✗	✗	✗	●	✗	✗	✗
Delta-view (stress – baseline)	✓	✗	✗	✗	✗	✗	✗	✗	✗
Indian market + costs	✓	✗	✗	✗	✗	✗	✗	✓	●
AI / natural-language scenario	✗	✗	✗	✗	●	✗	✗	✓	●
Shareable PDF report	●	✓	✓	✓	●	✓	✓	●	✗

† = exists in the TradeVed *backtester* flow ([engine/validation.py](#)), but NOT yet wired into the *stress tester*. This is the single most important correction to make when reading this doc: the walk-forward/OOS *engine is already built and battle-tested* — it just needs to be surfaced into the stress/robustness flow.

Read of the matrix: TradeVed *uniquely* owns the bottom-left cluster (streaming MC, canvas spaghetti, delta view, Indian costs), is competitive on the scenario library, and **already has the walk-forward/OOS engine** competitors charge thousands for. Where the *stress tester* is thin is the rest of the middle band — trade-level MC, permutation p-values, overfitting probability (PBO/DSR), parameter heatmaps, tail metrics, and a single robustness score. That band is exactly where trust and differentiation live, and TradeVed is closer to it than the matrix first suggests because the hardest piece (walk-forward) is done.

4. Most Loved Features Across the Market

What users repeatedly praise (Reddit r/algorithmtrading, product reviews, forums, YouTube):

- Instant visual feedback** — signals overlaid on price; equity curve that updates as you tweak. *"The visual component is perfect for understanding why a trade was triggered."* (TradingView's #1 loved trait.) → **TradeVed already nails this with the live canvas.**
- Monte Carlo that exposes hidden drawdown** — Build Alpha's most-cited example: a backtest showing \$1,663 max DD revealed a **\$5,195 potential DD** under MC. Traders *love* being shown the drawdown they'd otherwise discover live.
- A single number / letter grade** — BacktestBase's "30-point score" and grade is beloved because it collapses a wall of stats into a decision. People trust and *share* a grade.
- Walk-forward "it survived out-of-sample"** — the single most trusted robustness signal among serious quants. AmiBroker/Build Alpha/Wealth-Lab users treat passing WFO as the bar for going live.
- Parameter-sensitivity heatmaps / 3D surfaces** — QuantConnect's & Build Alpha's most-loved discovery tool: *"is my profit a robust plateau or a single lucky spike?"*
- "Scenario priced-in" tracking** — HiddenLevers' signature: showing a scenario is "50% priced in" as oil moves \$50→\$75→\$100. Advisors say it's the best client-trust tool they have.
- Variance / forward simulation** — Build Alpha's "how will this do over the next N trades at varying win rates" — users love a *forward* projection, not just a backward one.
- No-code + natural language** — AlgoTest's AI agent "920" and TrendSpider's AI Lab are loved for removing the coding wall entirely.
- Closing the loop** — TradeZella's auto-feed of backtest → live → analytics. The #1 complaint about *every other tool* is that it "stops at the backtest."
- Speed at scale** — vectorbt (1M orders in ~70ms) and QuantConnect cloud parallelism let users test thousands of variants; speed itself is a loved feature.

5. Most Requested / Missing Features

The recurring wishlist and frustrations across communities:

- **"Tell me if it's overfit."** The #1 anxiety in algo trading. Users want an explicit overfit verdict, not just good-looking numbers. (Red flags they cite: too-high returns, too-low DD, too many tuned parameters.)
- **"Close the gap between backtest and live."** Universally cited: *"Most tools stop at the backtest and you're on your own."* People want shadow/paper comparison and a realism model (spread, commissions, slippage, partial fills, borrow/funding — and Build Alpha-style **"then double the costs"**).
- **Realistic execution** — delayed fills, queue position, partial fills, slippage that scales with size. Backtests assuming perfect fills are distrusted.
- **Statistical significance (a real p-value)** — *"would this look this good on data with no edge?"* MC permutation p-values are requested but rarely shipped.
- **A robustness/confidence score** — one decision-grade number with a clear pass bar.
- **Probability of ruin / risk of hitting a kill-threshold** — requested constantly by risk-conscious traders.
- **Tail metrics** — CVaR / Expected Shortfall, not just max drawdown.
- **Multi-asset / portfolio stress with correlation breakdown** — the thing institutions have and retail doesn't: *"in a crash, diversification disappears."*
- **Indian-market-specific scenarios** — circuit breakers, SEBI events, expiry-day gamma, demonetization-style shocks. *Nobody* serves these.
- **Shareable, beautiful reports** — PDF/HTML you can hand to a client, a prop desk, or post.

6. Institutional Practices

How professional quant funds, prop firms, and bank risk desks actually validate robustness — the practices worth importing into a retail product:

6.1 Validation methodology

- **Out-of-sample is sacred.** Optimize on in-sample, *never touch* the OOS slice until the end. Hold-out alone is considered weak; serious desks use **walk-forward** (rolling re-optimization, e.g., 9 sequential splits) and increasingly **Combinatorial Purged Cross-Validation (CPCV)** — which purges overlapping samples and embargoes around test windows to kill information leakage, and produces a *distribution* of OOS performance rather than a single path. CPCV demonstrably yields lower **Probability of Backtest Overfitting (PBO)** than K-fold or walk-forward.
- **Multiple-testing correction.** When you try many strategies/parameters, the "best" is partly luck. Desks deflate for this:
- **PBO via CSCV** (Bailey, Borwein, López de Prado, Zhu) — estimates the probability the selected config is overfit.
- **Deflated Sharpe Ratio (DSR)** — corrects the Sharpe for selection bias under multiple trials *and* non-normal returns.
- **Monte Carlo Permutation Test (MCPT)** — permute/shuffle the price series, re-run, and ask "would no-edge data have looked this good?" to get a real p-value.
- **Synthetic + historical dual validation.** Every model is tested on *both* real history and synthetic/noise-injected data to avoid relying on one-off anomalies.

6.2 Risk measurement

- **Expected Shortfall (CVaR) over VaR.** The Basel III FRTB replaced 99% VaR with **97.5% Expected Shortfall** as the primary market-risk measure — ES captures *how bad* the tail is, not just its frequency. Retail tools almost never show this.
- **Reverse stress testing.** Instead of "apply a 30% crash, see the loss," ask **"what scenario would break this strategy / blow the risk budget?"** and then characterize how plausible that is. Powerful and almost absent in retail.
- **Factor & macro stress.** Shock factors (rates, credit, vol, oil) and propagate through exposures; model **correlation breakdown** (correlations → 1 in panics, so diversification benefit evaporates — MSCI's "broken-arrow" test).

6.3 Pre-deployment discipline

- **Shadow / paper trading with production plumbing** — full logging, daily P&L reconciliation, real order-type logic, queue priority, latency — *before* a dollar is risked.
- **Cost realism then stress the costs** — model spread, commission, slippage, partial fills, borrow, funding — then **double the costs** and check the edge survives.
- **Kill-switches & live controls** — exposure limits, loss-cutting rules, auto-pause on regime shift or underperformance.

Takeaway for TradeVed: importing even *simplified, well-explained* versions of PBO, DSR, MCPT, CVaR/ES, walk-forward, and reverse stress testing would put a retail product on institutional footing — and none of TradeVed's retail competitors do this.

7. Where TradeVed Stands Today (Honest Self-Assessment)

From a line-level read of `engine/stress.py`:

Genuine strengths (keep & market these):

- **13 scenario presets** with methodologically sound *pre-strategy* OHLCV perturbation (`apply_stress` is a pure, deep-copying function — input never mutated). Scenarios include persistent crashes (`persist=True` prevents the snap-back/buy-low-sell-high artifact), bounces (GFC), V-recoveries (COVID), pump-dump, whipsaw/AR(1) chop, gap risk, vol spike, outlier injection.
- **Timeframe-aware shock durations** (`_candles_per_day` converts "days" → candles) so 1d and 4h runs are semantically comparable.
- **Monte Carlo with per-run magnitude jitter** (`severity × uniform(0.75, 1.25)`) + random shock start — paths fan out in timing *and* intensity (most retail MC only varies one).
- **Percentile aggregation** (P5/P50/P95/worst/best) for return, DD, Sharpe, Sortino, win-rate; an **equity fan** and up to **100 spaghetti paths** subsampled to 200 pts.
- **Live SSE streaming endpoint** (`/api/stress/stream`) emitting `baseline → run×N → complete`, with `asyncio.to_thread` so events flush — feeding a **canvas spaghetti chart** (1,000+ paths, HSL-colored, hover, click-to-pin) with a **delta-view toggle**. This UX is ahead of the entire retail field.
- **A full out-of-sample validation engine already exists** (`engine/validation.py`): `run_holdout` (train/test split + stable/degraded/failed verdict) and `run_walk_forward` (per-window grid-search by Sharpe → apply best params to OOS step → roll forward → aggregate OOS metrics + stitched validation curve). Exposed in the backtester via `validation_mode`, `wf_window`, `wf_step` and rendered in `ValidationPanel.tsx`. **This is a major asset most retail competitors lack — it's just not connected to the stress tester yet.**

Structural limitations:

- **Single-asset only** — no portfolio, no correlation breakdown.
- **Stress = data perturbation only** — no *trade-level* MC (reshuffle/resample), no parameter perturbation, no permutation p-value. (*Walk-forward/OOS exist but in the backtester flow — not yet reused by the stress tester to produce an overfit-aware robustness score.*)
- **Output = distribution of outcomes, not a verdict** — there's no robustness score, no overfit probability, no tail metric (CVaR/ruin), no statistical-significance test. The user still has to interpret.
- **Slippage stress is a single multiplier** (`slip_multiplier`) — no delayed/partial-fill modeling.
- **Scenarios are global price shocks**, not factor/macro-linked, and not Indian-specific.

Verdict: TradeVed has built the *hard, beautiful parts* — the scenario engine, the streaming visualization, and a **walk-forward/OOS validation engine** — that competitors either lack or charge thousands for. What's missing is *connecting* that validation science to the stress tester and distilling it into a trusted verdict (a robustness score, overfit probability, tail metrics). That's a very good position to be in: the remaining work is mostly **integration + scoring**, not a rewrite, because the most expensive component (walk-forward) already exists and works.

8. Product Gaps

Ranked by how much they hurt trust/differentiation:

1. **No single robustness verdict.** Output is a distribution; users want a grade + pass/fail. (Build Alpha, BacktestBase prove demand.)
2. **Overfitting science not connected to stress.** Walk-forward + OOS *exist* (`engine/validation.py`) but only in the backtester — the stress tester doesn't reuse them, and there's still no CPCV, PBO, DSR, or MCPT p-value anywhere. Wiring the existing walk-forward into a stress-side overfit signal is low-hanging fruit; the rest separates "trustworthy" from "toy."
3. **No tail-risk layer.** No CVaR/Expected Shortfall, no probability of ruin — both heavily requested.
4. **No trade-level Monte Carlo.** Only data-path MC. Reshuffle/resample MC (and trade-skip) is the most familiar MC to traders and cheap to add.
5. **No parameter-sensitivity heatmap.** The single most-loved robustness *discovery* tool (QuantConnect/Build Alpha) — and TradeVed has the optimizer data to build it.
6. **No execution-realism stress.** Delayed/partial fills, size-scaled slippage, "double the costs" toggle.
7. **No portfolio / correlation-breakdown stress.** The institutional crown jewel; nobody serves retail here.
8. **No reverse stress testing.** "What breaks me?" — high wow-factor, low competition.
9. **No Indian-specific scenarios.** A defensible moat given the existing Indian cost model.

10. No AI/natural-language scenario builder. AlgoTest/TrendSpider show this is now table-stakes-trending.
11. No shareable stress report (PDF). Blocks virality and pro/advisor use.
12. Doesn't close the loop to paper/live. The universal complaint.

9. Opportunities for TradeVed

Underserved segments

- **Indian retail algo traders** (exploding, served only by shallow tools like Streak/AlgoTest with *zero* stress testing). TradeVed's Indian cost model + F&O lots is a ready-made wedge.
- **Serious retail quants priced out of Build Alpha/desktop tools** who want robustness science in a *web* product with great UX.
- **Prop-firm-funded traders** who must *prove* robustness to keep an account — a robustness score + PDF report is exactly their need.

Cheap differentiators (you already own the building blocks)

- **Regime-aware Monte Carlo — the biggest easy win.** Your MC currently fans out with flat `severity × uniform(0.75, 1.25)` jitter (`stress.py:456`). You *already ship* `engine/regimes.py` (timeframe-aware bull/bear/sideways detection, run after every backtest). Feed it into the MC loop so each run samples **regime-specific volatility** — bear-regime runs draw fatter-tailed, higher-vol shocks; sideways runs draw tighter ones. BacktestBase only does this "partially"; with `regimes.py` already built, TradeVed can do it *better*, and it makes the spaghetti fan **physically realistic** instead of uniform noise. Low effort, high uniqueness.
- **No-lookahead "point-in-time" guarantee + badge.** QuantConnect's single most-trusted feature, and retail tools are widely distrusted here. Audit the signal path so indicators never peek at future candles, then display a "✓ No lookahead bias" badge on results. Cheap to verify, outsized trust payoff.
- **Selectable MC models.** Portfolio Visualizer's loved option — let the user choose the MC engine: **historical bootstrap**, **block bootstrap** (preserves autocorrelation), **parametric (μ/σ)**, or your existing **scenario-perturbation** method. One dropdown, several methods, broad appeal.

UX / visualization innovations (extend existing strengths)

- The canvas spaghetti chart is a unique asset — extend it to **walk-forward windows** (each OOS slice as a segment), **parameter heatmaps**, and **reverse-stress "break point" markers**.
- A "**Robustness Gauge**" hero component (A+ → F dial) at the top of `StressResults` — the single most shareable element you could add.
- "**Priced-in**" **scenario meter** (HiddenLevers-style) tied to live price vs scenario target.
- "**What-If**" **stressed-path overlay** — draw the perturbed price path *on top* of the real candlestick chart (LuxAlgo-style), distinct from the equity delta-view, so users *see* the shock applied to price.
- **Multi-strategy "book" stress** — stress all of a user's deployed strategies at once (HiddenLevers' household-level idea, applied to a book of algos) and aggregate the blast radius.

AI-assisted workflows

- **Natural-language scenario builder:** "crash NIFTY 30% over 2 months with two relief rallies, then stay down" → parameters for a `StressScenario` . (You already have the dataclass; this is a thin LLM-to-params layer.)
- **AI verdict narrator:** turn the score + metrics into a plain-English paragraph ("Your strategy survived 10/13 scenarios but is fragile to slow bleeds and shows HIGH overfit risk because OOS Sharpe is 38% of in-sample...").
- **Auto-recommendations:** "reduce parameter count," "widen the EMA plateau," "your edge vanishes after costs ×2."

Automation opportunities

- **Robustness gate** (StrategyQuant-style): in the optimizer (`crypto_optimizer.py` , `indian_futures_optimizer.py`), auto-reject configs that fail MC/walk-forward — surface only robust strategies.
- **Scheduled re-stress:** re-run the suite weekly on new data; alert if a deployed strategy's robustness score degrades.

10. Recommended Features — Ranked by Priority

Scored on **Impact** (trust + differentiation), **Effort**, **Demand**, **Moat**. P0 = do first.

#	Feature	Impact	Effort	Demand	Moat	Priority
1	Robustness Score (A+→F) + Gauge		Low		Med	P0
2	Probability of Ruin + CVaR/ES in MC output		Low		Med	P0
3	Trade-level MC (reshuffle/resample) + trade-skip %		Low		Low	P0
4	Indian-specific scenario presets		Low			P0
5	Wire EXISTING walk-forward/OOS into stress robustness score (reuse <code>validation.py</code>)		Low		Med	P0
6	Parameter-sensitivity heatmap		Med		Med	P1
7	MC permutation p-value (MCPT)		Med			P1
8	Execution-realism stress (delay/partial/×2 costs)		Med		Med	P1
9	AI verdict narrator + NL scenario builder		Med			P1
10	Deflated Sharpe + PBO (overfit engine)		High			P2
11	Reverse stress testing ("what breaks me?")		High			P2
12	Portfolio + correlation-breakdown stress		High			P2
13	Shareable PDF stress report		Low		Low	P1
14	CPCV engine		High			P3
15	Paper/live loop-closure comparison		High		Med	P3
16	Regime-aware MC (reuse <code>regimes.py</code> for per-run vol)		Low–Med			P1
17	No-lookahead audit + trust badge		Low		Med	P1
18	Selectable MC models (bootstrap / parametric)		Med		Low	P2
19	"What-If" stressed-path overlay on chart		Low		Med	P2

11. TradeVed Opportunity Roadmap

Phase 1 — "Trust Layer" (2–4 weeks) — *ship a verdict, not just a chart*

Turn the existing distribution into a decision.

- **Robustness Score + A-F Gauge** (\$13.1). Computed from data you *already* produce (`monte_carlo` percentiles + `baseline` / `stressed`). Hero dial in `StressResults.tsx` .
- **Tail metrics in MC aggregation**: add CVaR/Expected Shortfall (5%), Probability of Ruin, return P1, MC max-DD P95 to `_pcts()` / `aggregate_stress_results()` in `stress.py` .
- **Trade-level MC + trade-skip slider**: add `mc_mode: "data" | "trades" | "both"` and a `trade_skip_pct` (default 10%) — reshuffle/resample the realized trade list; cheap and familiar.
- **4 Indian scenario presets**: `demonetization_2016` , `covid_nifty_mar2020` , `yes_bank_2020` , `expiry_gamma_squeeze` — add to `SCENARIO_PRESETS` , `SCENARIO_DISPLAY/GROUPS/DEFAULTS` (StressSidebar) and `StressScenarioKey` (types.ts).
- **Wire the EXISTING walk-forward/OOS engine into the stress score** (*reuse, not build*): call `run_walk_forward` / `run_holdout` (already in `engine/validation.py`) from the stress flow and compute **Walk-Forward Efficiency (WFE)** = aggregated OOS Sharpe ÷ mean in-sample `train_sharpe` . This makes the Robustness Score **overfit-aware on day one** with no new engine — `run_walk_forward` already returns per-window `train_sharpe` and OOS `sharpe` .
- **Regime-aware MC** (*reuse* `regimes.py`): replace the flat `uniform(0.75, 1.25)` per-run jitter with **regime-conditioned volatility** drawn from the regime mix `classify_regimes` already detects — physically realistic fan-out, and a differentiator no retail tool does well.
- **No-lookahead badge**: audit the signal path, then show a "✓ No lookahead bias" trust badge on results.
- **AI verdict narrator**: one paragraph generated from score + metrics.

Outcome: TradeVed goes from "shows you stress paths" to "**grades your strategy and explains why**" — instantly differentiated from every retail competitor, and overfit-aware immediately by reusing code that already exists.

Phase 2 — "Validation Science" (4–8 weeks) — *earn quant trust*

- **Deepen the walk-forward already wired in Phase 1**: render per-window OOS segments on the canvas spaghetti; layer **CPCV-style multiple train/test splits** on top of the existing `run_walk_forward` to produce a *distribution* of OOS outcomes (not a single stitched path); use it as an auto-reject **gate** in `crypto_optimizer.py` / `indian_futures_optimizer.py` .
- **Parameter-sensitivity heatmap**: 2-param grid → 2D/3D surface; flag "robust plateau vs lucky spike." (Reuse optimizer output.)
- **MC permutation p-value (MCPT)**: permute price series, re-run, report p-value ("1.2% chance this is luck").
- **Execution-realism stress**: delayed-fill (1–2 candles), partial fills, size-scaled slippage, "**double the costs**" toggle.
- **Shareable PDF report** (extend `frontend/report.py`): score, scenario table, MC fan, heatmap, verdict.

Phase 3 — "Institutional Moat" (8–16 weeks) — *features no retail tool has*

- **Overfit engine**: Deflated Sharpe Ratio + PBO via CSCV → "Overfit Risk: LOW/MED/HIGH."
- **Reverse stress testing**: search for the minimal shock (depth/duration/vol) that pushes the strategy below a user threshold; mark the "break point."
- **Portfolio + correlation-breakdown stress**: multi-symbol; in crash scenarios force pairwise correlations toward 1.0 and show diversification collapse.
- **CPCV** for the most rigorous OOS distribution.
- (Stretch) **Paper/live loop-closure**: compare deployed P&L to the stressed envelope; alert on robustness decay.

12. The Ultimate Stress Tester — Blueprint

A layered design. TradeVed already owns Layers 1 and the Visualization layer; the blueprint adds 2–6.



Core feature set (must-have)

- Scenario library (historical + Indian + custom) — **HAVE, extend**
- Dual Monte Carlo: data-path (HAVE) + **trade-level** with skip %
- Walk-forward / OOS validation — **HAVE in backtester (validation.py); wire into stress**
- Tail risk: CVaR/ES + Probability of Ruin
- **Robustness Score (A→F)** with per-axis breakdown
- Live spaghetti + delta view — **HAVE**

Advanced feature set

- Parameter-sensitivity heatmap (robust-plateau detection)
- MC permutation p-value; Deflated Sharpe; PBO
- Reverse stress testing (break-point search)
- Execution realism (delay/partial fills, size-scaled slippage, ×2 costs)
- Portfolio + correlation-breakdown stress (+ multi-strategy "book" aggregation)
- **Regime-aware MC** (per-run vol from regimes.py) + **selectable MC models** (historical/block bootstrap, parametric, scenario-perturbation)
- **No-lookahead / point-in-time guarantee** with a visible trust badge

AI-powered capabilities

- **NL scenario builder** (text → StressScenario)
- **AI verdict narrator** (metrics → plain-English risk story + fixes)
- **Auto-robustness-gate** in optimizers (reject fragile configs)
- **Anomaly explainer** ("why did run #318 lose 60%? — the COVID crash hit during a cascade entry")

Validation engine (how it works)

Steps 1–3 already exist in `engine/validation.py` (`run_walk_forward` , `run_holdout`) and run in the backtester today. The net-new work is (a) calling them from the stress flow and (b) adding steps 4–6.

1. Split history → walk-forward windows (`run_walk_forward` , default window=252 / step=63) **or** CPCV groups (purge + embargo). (*walk-forward: HAVE*)
2. Re-optimize in-sample (grid-search best params by Sharpe), record OOS performance per window → **distribution**, not a point. (*HAVE*)
3. **WFE** = mean OOS Sharpe / mean IS Sharpe (≥ 0.5 healthy). (*computable now from `run_walk_forward` output*)
4. **PBO** via CSCV: fraction of combinatorial splits where the IS-best config underperforms median OOS. (*new*)
5. **MCPT**: permute prices, re-run N times → p-value = rank of real performance in the null. (*new*)
6. **DSR**: deflate Sharpe for #trials and non-normality. (*new*)

Reporting engine

- One-click PDF/HTML (extend `frontend/report.py`): score gauge, scenario Δ -table, MC fan, heatmap, WFO segments, tail metrics, AI verdict. Shareable link for prop desks/advisors/social.

Risk analysis layer

- CVaR/ES (5%), Probability of Ruin (P(equity < kill-threshold)), DD percentiles, recovery-time distribution, and (portfolio) correlation-breakdown impact.

13. Scoring Frameworks (Implementation Spec)

13.1 TradeVed Robustness Score (TRS) — 0–100 → A+...F

A weighted composite, **computable mostly from data** `stress.py` already returns. Each sub-score normalized 0–100.

Axis	Weight	Inputs (already available unless noted)	Intuition
Scenario Survival	30%	% of 13 scenarios with $\Delta\% \geq -X$; median $\Delta\%$; worst scenario $\Delta\%$	Does it survive known crises?
MC Stability	25%	MC <code>return_pct.p5 > 0?</code> ; P5/P50 ratio; dispersion (P95–P5)	Is the edge consistent across paths?
Tail Safety	20%	CVaR(5%); Probability of Ruin; MC <code>max_drawdown_pct.p95</code> (<i>add</i>)	How ugly is the worst case?
Overfit Resistance	25%	Walk-Forward Efficiency (<i>available NOW via <code>validation.py</code></i>); MCPT p-value & param-plateau width (<i>Phase 2 refinements</i>)	Will it survive <i>unseen</i> data?

$$\text{TRS} = 0.30 \cdot \text{Survival} + 0.25 \cdot \text{MC_Stability} + 0.20 \cdot \text{Tail_Safety} + 0.25 \cdot \text{Overfit_Resistance}$$

Grade: A+ ≥ 90 • A 80–89 • B 70–79 • C 60–69 • D 50–59 • F < 50

Because walk-forward **already exists**, the Overfit Resistance axis can use **WFE from day one** — so the full 4-axis TRS is achievable in Phase 1 (MCPT p-value and parameter-plateau width refine this axis later). Only fall back to a **Provisional TRS** (first three axes, re-weighted to 100%) when the user's date range is too short for walk-forward (`n < wf_window + wf_step`), and label it "Provisional — widen date range for the overfit-resistance score."

13.2 Overfit Risk (separate verdict)

```

Signals → verdict:
WFE  = OOS_Sharpe / IS_Sharpe      (<0.3 bad, 0.3-0.5 caution, >0.5 good)
PBO  = P(backtest overfit) via CSCV (>0.5 bad)
MCPT = permutation p-value        (>0.10 bad → result indistinguishable from luck)
DSR  = deflated Sharpe            (≤0 bad → not significant after multiple-testing)

Overfit Risk = LOW  (≥3 green) · MEDIUM (mixed) · HIGH (≥2 red)

```

13.3 Confidence Band (sample-aware)

Don't show a crisp score on thin evidence. Scale a confidence interval by **#trades**, **#MC runs**, and **OOS-window count**:

```

low trades (<30) OR low MC (<200) → "Low confidence — widen date range / raise MC runs"
else                               → show TRS ± half-width from MC P5/P95 dispersion

```

13.4 Probability of Ruin & CVaR (drop-in for `stress.py`)

- Add to the `_pcts()` consumers in `run_stress_backtest` / `aggregate_stress_results` :
- `cvar_5 = mean(returns[returns <= percentile(returns, 5)])`
- `prob_ruin = mean(final_equity < kill_threshold)` where `kill_threshold = capital × (1 - max_loss_tolerance)` (default 50%)
- `ret_p1 = percentile(returns, 1)` and `maxdd_p95` (already have the array).
- Surface in `monte_carlo` payload → render in `StressResults.tsx` MC panels.

14. Sources

Institutional / portfolio stress

- [BlackRock Aladdin — Power of Stress Testing](#)
- [Bloomberg PORT — Build custom stress scenarios](#)
- [MSCI — A Stress Test to Incorporate Correlation Breakdown](#)
- [HiddenLevers — Interactive Stress Testing · Capterra reviews](#)
- [Top 10 Financial Stress Testing Platforms — Cotocus](#)

Robustness / quant tools

- [Build Alpha — Robustness Testing Guide · Monte Carlo Permutation · Monte Carlo Simulation](#)
- [StrategyQuant X — Types of robustness tests · 5 Monte Carlo methods](#)
- [AmiBroker — Walk-forward · Features · Review](#)
- [QuantConnect Review — LuxAlgo · Backtesting docs](#)
- [Portfolio Visualizer — Monte Carlo](#)
- [Wealth-Lab vs MultiCharts · NinjaTrader vs MultiCharts · MultiCharts WFO](#)
- [vectorbt \(GitHub\) · VectorBT PRO features](#)
- [BacktestBase — MC Stress Testing](#)
- [LuxAlgo — Stress Testing Your Algo · Backtesting ranked for retail quants](#)

Retail / Indian / charting

- [AlgoTest — Best options backtesting India · AlgoTest](#)
- [TrendSpider review](#)
- [MetaTrader 5 Strategy Tester](#)
- [TradeZella — best backtesting software · Tradewell](#)

Overfitting / statistical science

- [The Probability of Backtest Overfitting \(Bailey, Borwein, López de Prado, Zhu\) · SSRN](#)
- [The Deflated Sharpe Ratio \(SSRN\)](#)
- [Combinatorial Purged Cross-Validation — Papers With Backtest · Purged cross-validation \(Wikipedia\) · Traditional Backtesting is Outdated — Use CPCV](#)

- Expected Shortfall / CVaR explained · Reverse stress testing & dynamic loss models (arXiv)
 - Human-in-the-Loop Quant: anti-overfitting workflow
-

Prepared as a living strategy document — update as features ship and the competitive field moves.